

1 Complete List of 54 Motifs Used in this experiment

The 54 motifs used in this study span several categories of drought-related regulatory elements: ABA-responsive elements (ABRE, ABRE_core, CE1, CE3, ABRE_like, ACGT_core), dehydration/cold responsive elements (DRE_CRT, DRE_full, LTRE, LTRE_like, CBF_DRE, ICE_box), MYB/MYC binding sites (MYB_core, MYB1AT, MYB2, MYBCORE, MYC), WRKY binding elements (WBOX, WBOX_core), GT elements (GT1, GT1_core), heat shock elements (HSE, HSE_core), ethylene responsive elements (GCC_box, ERE), light/circadian elements (GBOX, IBOX, CAAT_box), stress elements (STRE), wound/pathogen elements (JASE, TBOX), NAC binding sites (NAC_core, NACRS), auxin responsive elements (AuxRE, AuxRE_rev), root-specific motifs (ROOTMOTIF), general elements (AS1, TATA_box, CACTFTPPCA1), and common 4-mer regulatory elements (ACGT, CACG, CATG, CGTG, GCGT, AACA, CAAC, AAAG, GATA, TATA, CAAT, GCCG, CGCG, TGAC, GTAC) [1], [2], [3], [4], [5], [6].

Here we provide the detailed list of the cis-regulatory motifs used in this study and their cross-references to the PLACE [7] and PlantCARE [8] databases. Motif sequences were implemented in the Python pipeline as regular-expression (regex) patterns and scanned on both the forward strand and its reverse complement. The *Motif Key Name* column gives the identifier used in the analysis pipeline, while the *PLACE Key Name* column gives the closest corresponding database entry. PLACE is available at https://www.dna.affrc.go.jp/PLACE/place_dat.shtml, and PlantCARE is available at <https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>.

Table 1 ABA-responsive cis-regulatory motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
ABRE	ACGTG[GT][CT]	ABREATCONSENSUS	S000406	PLACE
ABRE_core	ACGTG	ABRELATERD1	S000414	PLACE
CE1	TGCCACCGG	ABRECE1HVA22	S000014	PLACE
CE3	ACGCGTGTC	CE3OSOSEM	S000282	PLACE
ABRE_like	[AC]ACGT[GC][CGT]	ABREZMRAB28_LIKE	S000133	PLACE
ACGT_core	ACGT	ACGTATERD1	S000415	PLACE

Note: Motif sequences were scanned on both the forward strand and its reverse complement. PLACE database available at https://www.dna.affrc.go.jp/PLACE/place_dat.shtml.

Table 2 Dehydration and low-temperature responsive motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
DRE_CRT	CCGAC	LTRECOREATCOR15	S000153	PLACE
DRE_full	[AG]CCGAC	DRECRTCOREAT	S000418	PLACE
LTRE	CCGAC	LTRECOREATCOR15	S000153	PLACE
LTRE_like	ACCGACA	LTREATLTI78	S000157	PLACE
CBF_DRE	GTCGAC	CRTDREHVCBF2	S000411	PLACE

Note: DRE_CRT and LTRE share the same core sequence CCGAC but differ in their associated biological context and literature origin.

Table 3 MYB transcription factor binding motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
MYB_core	C[ACGT]GTT[AG]	MYBCORE	S000176	PLACE
MYB1AT	[AT]AACCA	MYB1AT	S000408	PLACE
MYB2	TAACTG	MYB2AT	S000177	PLACE
MYBCORE	C[ACGT]GTT[AG]	MYBCORE	S000176	PLACE

Note: MYB_core and MYBCORE share identical regex patterns and are retained as separate pipeline identifiers for compatibility with independent annotation tracks.

Table 4 MYC and E-box related motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
MYC	CA[ACGT][ACGT]TG	MYCCONSENSUSAT	S000407	PLACE
ICE_box	CA[ACGT][ACGT]TG	MYCCONSENSUSAT_ICE1	S000407	PLACE
NACRS	CATGTG	MYCATERD1	S000413	PLACE

Note: MYC and ICE_box share the same consensus sequence but differ in their associated transcription factor context; ICE_box is specifically linked to ICE1-mediated cold-stress signalling.

Table 5 WRKY and GT transcription factor binding motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
WBOX	TTGAC[CT]	ELRECOREPCRP1	S000142	PLACE
WBOX_core	TGAC	ELRECOREPCRP1_CORE	S000142	PLACE
GT1	G[AG][AT]AA[AT]	GT1CONSENSUS	S000198	PLACE
GT1_core	GAAAAA	GT1GMSCAM4	S000453	PLACE

Note: WBOX_core (TGAC) is the minimal WRKY-binding tetranucleotide contained within the full WBOX pattern (TTGAC[CT]).

Table 6 General regulatory and promoter motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
HSE	GAA[ACGT][ACGT]TTC	HSE	S000054	PLACE
HSE_core	[ACGT]GAA[ACGT]	HSE_CORE	S000054	PLACE
GCC_box	GCCGCC	GCCCORE	S000430	PLACE
ERE	A[AT]TTCAAA	ERELEE4	S000037	PLACE
GBOX	CACGTG	CACGTGMOTIF	S000042	PLACE
IBOX	GATAAG	IBOX	S000124	PLACE
CAAT_box	CCAAT	CCAATBOX1	S000030	PLACE
TATA_box	TATAAA	TATABOX2	S000132	PLACE
STRE	AGGGG	STRE	PlantCARE	PlantCARE

Note: STRE is sourced from PlantCARE only. PlantCARE is available at <https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>.

Table 7 Additional regulatory motifs including NAC, auxin, and root-related elements used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
JASE	CGTCA	JASE1ATOPR1	S000388	PLACE
TBOX	ACTTTG	TBOX	PlantCARE	PlantCARE
NAC_core	CACG	NACCORE_CACG	S000042	PLACE
AuxRE	TGTCTC	ARFAT	S000270	PLACE
AuxRE_rev	GAGACA	ARFAT_REVCOMP	PlantCARE	PlantCARE
ROOTMOTIF	ATATT	AT1BOX_CORE	S000025	PLACE
AS1	TGACG	AUXRETGA1GMGH3_CORE	S000234	PLACE
CACTFTPPCA1	[CT]ACT	CACTFTPPCA1	S000449	PLACE

Note: AuxRE_rev and TBOX are sourced from PlantCARE only. All other motifs in this table are cross-referenced with PLACE.

Table 8 Simple 4-mer regulatory sequence features used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
ACGT	ACGT	ACGTATERD1_4MER	S000415	PLACE
CACG	CACG	CACGTGMOTIF_CORE	S000042	PLACE
CATG	CATG	MYCATERD1_CORE	S000413	PLACE
CGTG	CGTG	ABRE_CGTG_CORE	S000406	PLACE
GCGT	GCGT	CE3_GCGT_CORE	S000282	PLACE
AACA	AACA	MYBGAHV_CORE	S000181	PLACE
CAAC	CAAC	CAAC_PLANTCARE	PlantCARE	PlantCARE
AAAG	AAAG	DOFCOREZM	S000265	PLACE
GATA	GATA	GATABOX	S000039	PLACE
TATA	TATA	TATABOX2_CORE	S000132	PLACE
CAAT	CAAT	CCAATBOX1_CORE	S000030	PLACE
GCCG	GCCG	AGCBOXNPGLB_CORE	S000232	PLACE
CGCG	CGCG	CGCGBOXAT_CORE	S000501	PLACE
TGAC	TGAC	ELRECOREPCR1_CORE	S000142	PLACE
GTAC	GTAC	CURECORECR	S000493	PLACE

Note: CAAC is sourced from PlantCARE only. All remaining 4-mer features are cross-referenced with PLACE. Each 4-mer was scanned on both the forward strand and its reverse complement.

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